

1. Introduction

1.1 Introduction

(1) What is Heat Transfer (HT)?

- Thermal energy transport in space and time.
- "Heat transfer is energy in transit due to temperature difference."
- Heat transfer supplements the first and second laws of thermodynamics which deal with equilibrium phenomena. HT gives information on the "rate" of thermal energy transport.
- 1st law $\Rightarrow$ Energy is conserved.
- 2nd law $\Rightarrow$ Thermal energy is transferred from the high-temperature region to low temperature region.
- HT $\Rightarrow$ How fast? In what way (e.g., spatial distribution of thermal energy)?

(2) Why study HT?

- HT (thermal energy transport) is central to numerous engineering and scientific problems.
- ME: power system, automotive engineering
- EE: electronic packaging, thermal load control
- ChE: reaction control
- Materials E: material synthesis

(3) What are thermal energy and temperature (T°)?

- Thermal energy: atomic/molecular/electronic kinetic energy
- Temperature:
 - Measure to determine how hot/cold a material is (intensity of thermal energy)
 - Criterion to determine the direction of thermal-energy transport
 - From a microscopic view, temperature represents atomic or molecular kinetic energy (translation / vibration / rotation)

(4) What carries (transfers) thermal energy?

- Particles: atoms, molecules, electrons
- Phonon (atomic vibration)
- Photon (electromagnetic radiation)

[Symbols and units]

- Thermal energy: $E=[J]$ (thermal energy or heat has the same unit as work (=force×displacement))
- Heat transfer rate: $q=[W]=[J/s]$
- Heat transfer rate per unit length $q' = q/L = [W/m]$
- Heat transfer rate per unit length $q'' = q/A = [W/m^2]$
- Temperature: $T=[^{\circ}C]$ or $[K]$ $T(^{\circ}C) = T(K) + 273.15$

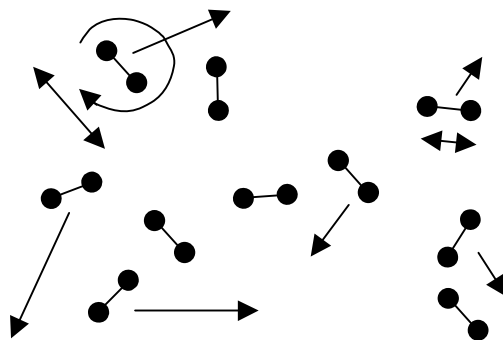
Note: When $^{\circ}C$ or K unit is in the denominator, unit change does not affect the numerical value (e.g., specific heat C_p $1 J/kg \cdot ^{\circ}C = 1 J/kg \cdot K$, thermal conductivity k $1 W/m \cdot ^{\circ}C = 1 W/m \cdot K$)

1.2 Basic modes of HT

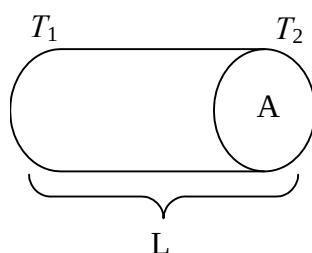
(1) Conduction (Heat diffusion)

HT due to random molecular (atomic) motion in a solid or stationary fluid
(HT in a solid)

Solids and stationary liquids are macroscopically motionless. However, molecules, atoms (lattices), electrons, etc. are constantly moving through translation, vibration, and rotation in a random direction. The random motion diffuses from high temperature to low temperature region until thermal equilibrium is reached. Molecular random motion is dominant in stationary fluids while lattice vibration and motion of free electrons are important in the cases of solid.



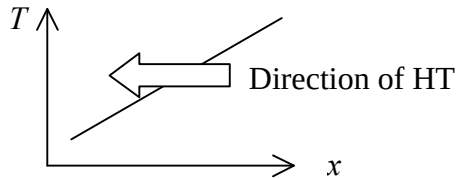
Rate equation: Fourier's law



$$q \sim (\text{area}) \times (\text{Temperature difference}) / (\text{distance}) = A \frac{\Delta T}{\Delta x} = A \frac{|T_1 - T_2|}{L}$$

In 1-d, HT rate $\boxed{q = -kA \frac{dT}{dx}}$. (k : thermal conductivity), HT rate per unit area $q'' = -k \frac{dT}{dx}$

- HT rate is proportional to the "temperature gradient" ($\cong$ temperature difference/distance)
- Minus sign (-) denotes the direction of heat flow.



[Analogy in transport phenomena]

Note that the diffusion mechanism is not only for HT but also for momentum/mass transfer!

Quantity		Driving potential	Macroscopic results	Physical model
Momentum	in transit due to	Velocity gradient	μ (viscosity)	$\tau = \mu \frac{du}{dx}$ (Shear law)
Energy		Temperature gradient	K (thermal conductivity)	$q'' = -k \frac{dT}{dx}$ (Fourier's law)
Mass		Concentration gradient	D_{ij} (diffusion coefficient)	$j_a = -\rho D_{ab} \frac{dm_a}{dx}$ (Fick's law)

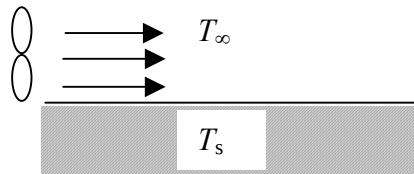
(2) Convection

HT with "bulk" or "macroscopic" motion of fluid

(HT between a solid object and fluid or HT in a moving fluid)

Temperature field is not determined only by the random molecular motion but also by the thermal energy carried by the bulk fluid motion.

Convection = Conduction + Bulk energy transport (advection)



- Convection HT includes the conduction HT mechanism. In the above case, the convection HT mechanism becomes identical to the conduction mechanism because there is no fluid motion on the wall (no slip BC).
- Bulk fluid flow can be either externally imposed (forced convection) or self-induced by buoyancy (natural or free convection).

Rate equation: Newton's law of cooling

$q \sim (\text{area}) \times (\text{Temperature difference}) \times (\text{fluid properties, flow speed, etc}) =$

$$A\Delta T = A(T_s - T_\infty)$$

$$q = hA(T_s - T_\infty) \quad (h: \text{convection heat transfer coefficient}), \quad q'' = h(T_s - T_\infty)$$

Table 1.1

HT Mode	h (W/m ² ·K)
Free convection	
Gases	2-25
Liquids	50-1,000
Forced convection	
Gases	25-250
Liquids	50-20,000
Convection with phase change (boiling, condensation)	2,500-100,000

- HT in liquid > HT in gas (material property, $k_{\text{liq}} > k_{\text{gas}}$, etc)
Aluminum: $k=202$ W/m·K, water: 0.6, air: 0.026
- Effect of latent heat critical

(3) Radiation

Energy transportation by EM (electromagnetic) wave

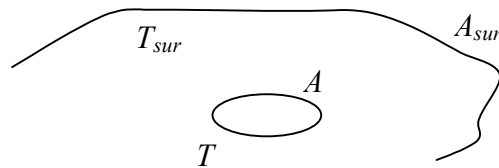
Thermal energy is emitted by EM wave (light) due to the changes in electron configurations of matter. The EM wave is always emitted from all surfaces at temperatures greater than absolute 0 K.

- No medium required
- Important in high-temperature HT, solar energy utilization, space applications
- EM wave can be visible (light) at high temperature

Rate equation: Stefan-Boltzmann law

Energy emitted by a surface per unit area (emissive power) $\sim T^4$

The magnitude of emissive power depends on material (surface). It has been known that the maximum value of thermal radiation at a given temperature T is $E_b = \sigma T^4$ (Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$). The imaginary surface having the maximum emission is called a blackbody. In the case of a real surface, $E = \epsilon \sigma T^4$ (ϵ : emissivity 0~1).



When a small object with an emissivity ϵ is surrounded by large surroundings at T_{sur} , the net radiation HT from the object to the surroundings is $q'' = \epsilon \sigma T^4 - \alpha \sigma T_{sur}^4$. If $A \ll A_{sur}$, the large surroundings can be treated as a blackbody. It has also been assumed that $\epsilon \approx \alpha$ (good emitter = good absorber). Therefore, $q = \epsilon \sigma A (T^4 - T_{sur}^4)$. $q'' = \epsilon \sigma (T^4 - T_{sur}^4)$

- HT is proportional to the fourth power
- Emissivity is a material (optical) property. If $\epsilon = 1$, the object is called a blackbody. For a blackbody, the emissive power is maximum $E_b = \sigma T^4$.

Summary

Conduction: $q = -kA \frac{dT}{dx}$

Convection: $q = hA(T_s - T_\infty)$

Radiation: $q = \epsilon \sigma A (T^4 - T_{sur}^4)$